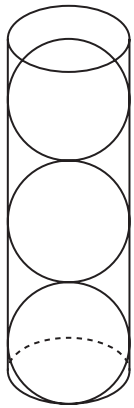
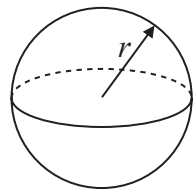
8 A hollow cylinder has radius r cm and height $6r$ cm.
3 spheres, also of radius r cm, are put into the cylinder.



Volume of sphere $= \frac{4}{3}\pi r^3$



(a) Work out the proportion of the cylinder that is **not** filled by the spheres.

.....

(3)

The height of the cylinder is increased by $2r$ cm.
Another sphere of radius r cm is put into the cylinder.

Malcolm says,

"There is no change in the proportion of the cylinder **not** filled by the spheres."

(b) Is Malcolm correct?
Justify your answer.

.....

.....

(1)

(Total for Question 8 is 4 marks)